🔬 MatMoE-Agent 综合科研报告

> **生成时间**: 2026-03-27 15:12:54 **用户指令**: 我想研发一款钙钛矿-硅两端叠层电池。为了图省事，我准备直接拿目前最成熟的 MAPbI3 盖在晶硅底电池上。这个器件设计方案合理吗？如果不合理，我该怎么改进钙钛矿层？

📑 最终研究报告 (Final Report)

优化后性能预测：

- **叠层效率：39.84%**（相比原方案提升10个百分点）
- 电流匹配度：0.72 mA/cm^{2}失配（仅2.6%失配率）
- 开路电压：2.277 V

四、具体实施建议

1. 材料配方推荐

$MAPb(I_{0.4}Br_{0.6})_3$ - 预计带隙：~1.75 eV - 合成方法：可通过MAI、PbI_2、$PbBr_2$前驱体溶液共混

2. 稳定性改进

- **A位混合：** $MA_{0.8}Cs_{0.2}$混合提高热稳定性
- **封装要求：** 必须严格隔绝水分和氧气
- **界面工程：** 需要优化钙钛矿/硅的界面能级对齐

3. 无铅替代方案（长期考虑）

- **Sn基钙钛矿：** $MASnI_3$带隙~1.3 eV（过小）
- **Ge基钙钛矿：** $MAGeI_3$带隙可调
- **双钙钛矿：** $Cs_2AgBiBr_6$带隙~1.8-2.0 eV

五、结论

原方案不合理之处： 1. **严重电流失配**（36%失配率） 2. **带隙组合非最优**（1.59 eV + 1.12 eV） 3. **铅毒性问题**限制商业化 4. **热力学不稳定**影响器件寿命

改进方案优势： 1. **带隙优化至1.75 eV**可实现近完美电流匹配 2. **理论效率达39.84%**，突破单结极限 3. **卤素合金化**工艺成熟，易于实施 4. **可进一步优化**A位阳离子提高稳定性

最终建议： 放弃直接使用$MAPbI_3$，改为开发带隙约1.75 eV的溴碘混合钙钛矿（如$MAPb(I_{0.4}Br_{0.6})_3$），并考虑A位Cs掺杂提高稳定性，同时积极研发无铅替代材料以满足环保要求。

🧠 智能体工具调用轨迹 (Agent Trace)

步骤 1: `tool_structure_search`

输入参数 (Action Input):

```
{
  "formula": "MAPbI3"
}
```

工具返回结果 (Observation):

```
D:\Workspace\MatMoE\MatAgent\cache\cif_cache\H6PbCI3N_mp-995214.cif
```

步骤 2: `tool_rag_search`

输入参数 (Action Input):

```
{
  "query": "MAPbI3 甲胺铅碘钙钛矿 带隙 bandgap 直接带隙 间接带隙 数值"
}
```

工具返回结果 (Observation):

```
----------【文献 1】(来源: 10.1039_d0ra03034a.md)
| $MABiI_3$          | 0.00          | 0.00      | _                                      |
the VBM at point R and a CBM and point Z. MAZnI3 formed a direct band gap of 1.89 eV at po

【文献 2】(来源: 10.1007_s40820-020-00457-7.md)
of Rh3+ is near the octahedral ). From frst-principle calculation, the bandgap values of M

【文献 3】(来源: 10.1038_s41598-020-79291-1.md)
|--------------|-------------------------|---------------------|-------------------------
| CH(NH2)2PbI3 | 3.98                    | 4.8                 | 5.2×10－7
bandgap of 1.46 eV, which is in a good agreement with the previously reported literature v

【文献 4】(来源: 10.1039_d4ra01640h.md)
| $MAPb(I_{0.58}Br_{0.42})_3$       | 8.20                        | 0.898             |
| $MAPb(I_{0.41}Br_{0.59})_3$       | 6.35                        | 0.834             |
| $MAPb(I_{0,28}Br_{0,72})_3$       | 3.18                        | 0.940             |
| $MAPb(I_{0.05}Br_{0.95})_3$       | 2.38                        | 0.832             |
the X-site iodine in the structure of MAPbI3.120 They effectively modified the band gap of

【文献 5】(来源: 10.1039_d1ra04520b.md)
The band gap energies were determined to be 1.41 eV, 1.40 eV, 1.46 eV, 1.42 eV, 1.43 eV, a

【文献 6】(来源: 10.3390_en15176326.md)
| Parameters                                                                  | FTO   | M
|-----------------------------------------------------------------------------|---------
| Band gap energy $E_g$ (eV)                                                  | 3.5

【文献 7】(来源: 10.1002_aesr.202200160.md)
value gradually decreases. The bandgap ranges of MAPbX3, FAPbX3 (1.48－3.02 eV), and CsPbX

【文献 8】(来源: 10.1039_d0ra03034a.md)
Most transition metal monoreplacements didn't form any band gaps. However, MACuI3 formed a
| Material           | Fund.(eV) | Opt. (eV) | PBE prev. (eV)                        | Exp
|--------------------|---------------|-----------|---------------------------------------|
| MAPbI 3 | 1.59          | 1.59      | 1.63, 1 1.59 17 | 1.55 18     |
| MASiI 3 | 1.70          | 1.70      | 0.14                                  | _
| MAVI 3  | 0.00          | 0.00      | 0                                     | _
| MACrI 3 | 0.00          | 0.00      | _                                     | _
| $MAMnI_3$          | 0.00          | 0.00      | _                                     |
----------
```

步骤 3: `tool_rag_search`

输入参数 (Action Input):

```
{
  "query": "硅 silicon 带隙 bandgap 间接带隙 直接带隙 数值"
}
```

工具返回结果 (Observation):

```
----------------------------------------【文献 1】(来源: 10.31026_j.eng.2025.10.04.md)
| G (W/m²) | Isc (A) | Voc(V) | Imax (A) | Vmax (V) | Pmax(W) | FF    | η     |
|----------|---------|----------|----------|----------|----------|-------|------|
| 250      | 2.25    | 44.20    | 1.890    | 37.50    | 70.9     | 0.713 | 14.6 |
| 500      | 4.73    | 45.60    | 4.500    | 38.98    | 175.3    | 0.812 | 18.1 |
| 750      | 7.18    | 46.00    | 6.32     | 39.56    | 250.1    | 0.759 | 17.2 |
| 1000     | 9.14    | 45.80    | 7.92     | 40.91    | 324.0    | 0.722 | 16.7 |
Radiative recombination losses (L3rad) are minimal, ranging from 0.12% to 0.44%, confirmin

【文献 2】(来源: 10.54254_2755-2721_24_20230693.md)
According to a research published by Zhao Peng et al., since the bandgap of crystalline si

【文献 3】(来源: 10.21275_sr24206001809.md)
【来源文档】: 10.21275_sr24206001809.md
【所属章节】: **7. Intrinsic Silicon Carbide Properties**
【正文内容】:
# **7. Intrinsic Silicon Carbide Properties**
As an intrinsic wide - bandgap semiconductor, silicon carbide has a bandgap energy of 3.2e
The wider bandgap results in a heavier effective mass for charge carriers and lower carrie

【文献 4】(来源: 10.54097_0yq1qy86.md)
Currently available commercial deep ultraviolet (DUV) devices, mainly photodetectors, stil

【文献 5】(来源: 10.1038_s41598-021-92443-1.md)
|            |      | DFT method             |         |      |                |
|------------|------|------------------------|---------|------|----------------|----------
| Material   | GGA  | GGA+U                  | GGA+SOC | mBJ  | GGA+mBJ+SOC    | Experimen
| Cs2CuSbCl6 | 0.61 | 0.61                   | 0.59    | 1.00 | 1.55           | 1.6614
| Cs2CuBiCl6 | 0.89 | 0.89                   | 0.85    | 1.20 | 2.00           |
|            |      | 0.8316, 0.9318, 2.0020 |         |      | -              | -

【文献 6】(来源: 10.1039_d0ra01764g.md)
Among, these materials only Cs2LiGaBr6 and Cs2NaGaBr6 have band gap values 0.731 eV and 0.

【文献 7】(来源: 10.1039_d4ra01640h.md)
treatment. The pressure-induced transitions from indirect to direct bandgaps represent an

【文献 8】(来源: 10.26418_elkha.v15i1.61351.md)
```

```
| Band gap (eV)
| Effective cond. band density (cm.5)
----------------------------------------
```

步骤 4: `tool_tandem_current_matcher`

输入参数 (Action Input):

```
{
  "bandgap_top_ev": 1.59,
  "bandgap_bot_ev": 1.12
}
```

工具返回结果 (Observation):


```
{'bandgap_combination_eV': {'top': 1.59, 'bottom': 1.12}, 'subcell_Voc_V': {'top': 1.293,
```


步骤 5: `tool_property_calculation`

输入参数 (Action Input):

```
{
  "file_path": "D:\\Workspace\\MatMoE\\MatAgent\\cache\\cif_cache\\H6PbCI3N_mp-995214.cif"
  "task": "formation"
}
```

工具返回结果 (Observation):


```
{'composition': 'H24 Pb4 C4 I12 N4', 'total_energy_eV': -4.381, 'formation_energy_eV_per_a
```


步骤 6: `tool_pv_performance_calc`

输入参数 (Action Input):

```
{
  "bandgap_ev": 1.59
}
```

工具返回结果 (Observation):

```
{'bandgap_input_eV': 1.59, 'Jsc_mA_cm2': 32.66, 'Voc_V': 1.325, 'FF_percent': 80.0, 'PCE_p
```

步骤 7: `tool_slme_efficiency_calculator`

输入参数 (Action Input):

```
{
  "bandgap_ev": 1.59,
  "direct_bandgap_ev": 1.59,
  "thickness_um": 2.0
}
```

工具返回结果 (Observation):

```
{'bandgap_fundamental_eV': 1.59, 'bandgap_direct_eV': 1.59, 'material_type': '直接带隙', '
```

步骤 8: `tool_goldschmidt_tolerance`

输入参数 (Action Input):

```
{
  "a_ion": "MA",
  "b_ion": "Pb",
  "x_ion": "I"
}
```

工具返回结果 (Observation):

```
{'target_system': 'MaPbI3', 'calculated_indices': {'tolerance_factor_t': 1.0163, 'octahedr
```

步骤 9: `tool_commercial_assessment`

输入参数 (Action Input):

```
{
  "formula": "MAPbI3"
}
```

工具返回结果 (Observation):


```
{'target_formula': 'MAPbI3', 'estimated_cost_usd_per_kg': 29.14, 'cost_breakdown': {'MA':
```


步骤 10: `tool_tandem_current_matcher`

输入参数 (Action Input):

```
{
  "bandgap_top_ev": 1.75,
  "bandgap_bot_ev": 1.12
}
```

工具返回结果 (Observation):


```
{'bandgap_combination_eV': {'top': 1.75, 'bottom': 1.12}, 'subcell_Voc_V': {'top': 1.443,
```